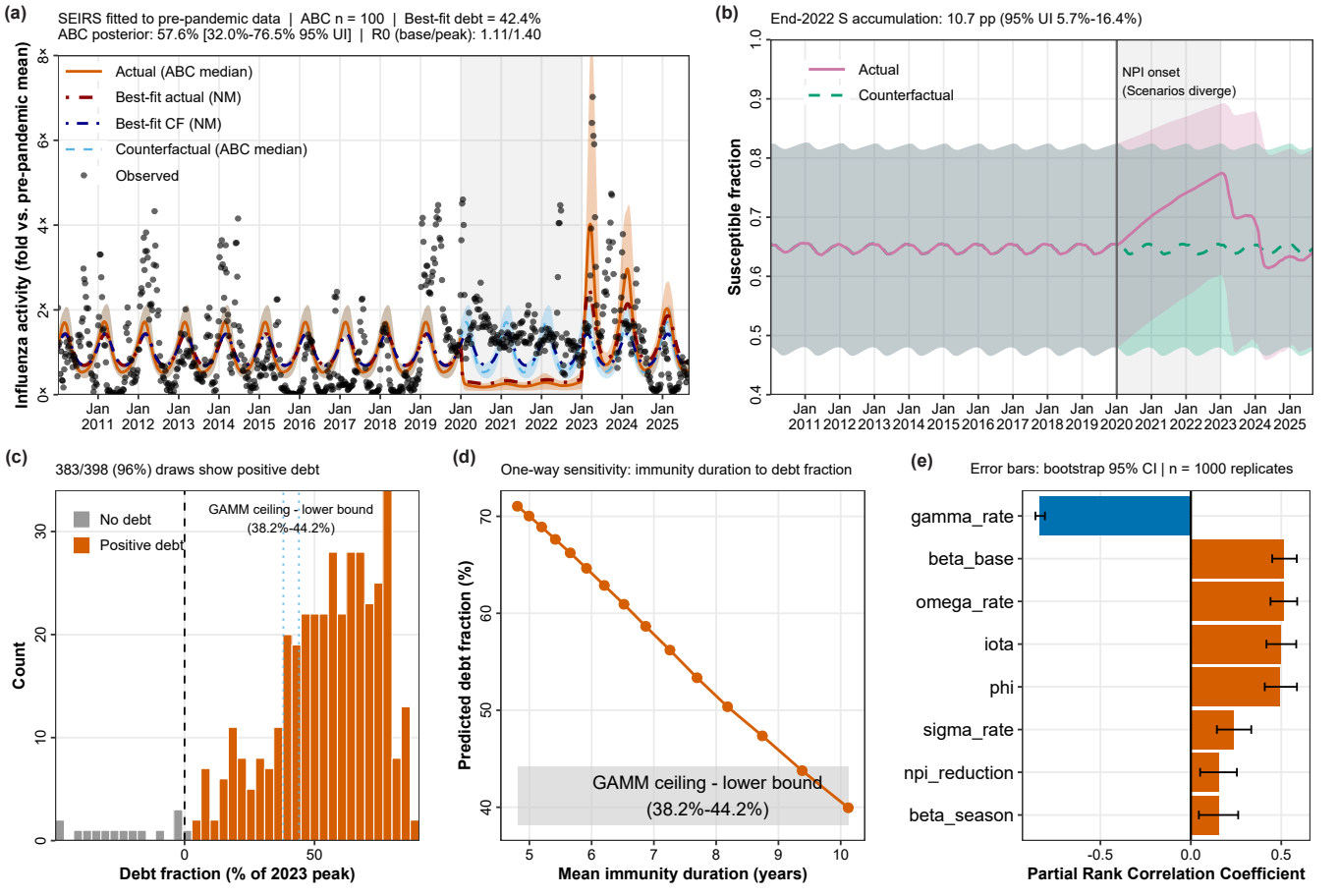

(a)
SEIRS fitted to pre-pandemic data | ABC n = 100 | Best-fit debt = 42.4%
ABC posterior: 57.6% [32.0%-76.5% 95% UI] | R0 (base/peak): 1.11/1.40
Actual (ABC median)
Best-fit actual (NM)
Best-fit CF (NM)
Counterfactual (ABC median)
Observed
Influenza activity (fold vs. pre-pandemic mean)
0×
2×
4×
6×
8×
Jan 2011
Jan 2012
Jan 2013
Jan 2014
Jan 2015
Jan 2016
Jan 2017
Jan 2018
Jan 2019
Jan 2020
Jan 2021
Jan 2022
Jan 2023
Jan 2024
Jan 2025
(b)
End-2022 S accumulation: 10.7 pp (95% UI 5.7%-16.4%)
Actual
Counterfactual
NPI onset
(Scenarios diverge)
Susceptible fraction
0.4
0.5
0.6
0.7
0.8
0.9
1.0
(c)
383/398 (96%) draws show positive debt
No debt
Positive debt
GAMM ceiling - lower bound
(38.2%-44.2%)
Count
0
10
20
30
0
50
Debt fraction (% of 2023 peak)
(d)
One-way sensitivity: immunity duration to debt fraction
Predicted debt fraction (%)
40
50
60
70
GAMM ceiling - lower bound
(38.2%-44.2%)
5
6
7
8
9
10
Mean immunity duration (years)
(e)
Error bars: bootstrap 95% CI | n = 1000 replicates
gamma_rate
beta_base
omega_rate
iota
phi
sigma_rate
npi_reduction
beta_season
-0.5
0.0
0.5
Partial Rank Correlation Coefficient